INCH-POUND

MS51819G
30 March 2022
SUPERSEDING
MS51819F
w/AMENDMENT 2
21 November 2019

DETAIL SPECIFICATION SHEET

ADAPTER, STRAIGHT, PIPE TO TUBE
MALE PIPE END, FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.

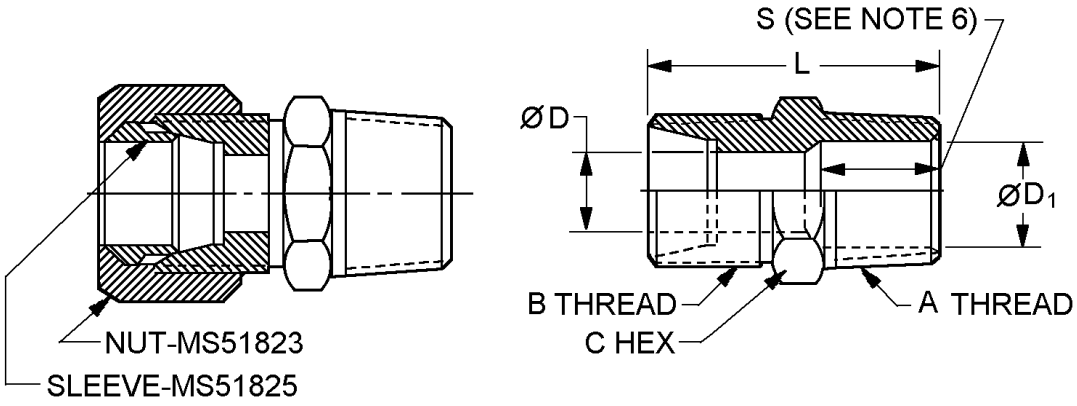


FIGURE 1. Adapter pipe to tube.

AMSC N/A
FSC 4730

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Dash number		Tube OD Nom inches (mm)	A Thread NPTF (see note 10)	B (see note 9)	C Hex flat Nom (see note 7)
Assembly	Body (see note 5)				
-1	-31	.1250 (3.175)	.125 -27	.3125 - 24UNF	7/16
-2	-32	.1875 (4.763)	.125 -27	.3750 - 24UNF	7/16
-3	-33	.2500 (6.350)	.125 -27	.4375 - 20UNF	1/2
-4	-34	.3125 (7.936)	.125 -27	.5000 - 20UNF	9/16
-7	-37	.3750 (9.525)	.250 - 18	.5625 - 18UNF	5/8
-9	-39	.5000 (12.700)	.375 - 18	.7500 - 16UNF	13/16
-12	-42	.6250 (15.875)	.500 - 14	.8750 - 14UNF	15/16
-14	-44	.7500 (19.050)	.750 - 14	1.0625 - 12UN	1 1/8
-15	-45	.8750 (22.225)	.750 - 14	1.1875 - 12UN	1 1/4
-18	-48	1.0000 (25.400)	1.000 - 11 1/2	1.3125 - 12UN	1 3/8
-20	-50	1.2500 (31.750)	1.250 - 11 1/2	1.6250 - 12UN	1 11/16
-21	-51	1.5000 (38.100)	1.500 - 11 1/2	1.8750 - 12UN	2
-22	-52	2.0000 (50.800)	2.000 - 11 1/2	2.5000 - 12UN	2 5/8

Dash number		D dia. Drill	D_1 dia. Drill	L inches (mm) ±.02 (0.51)	S max inches (mm)
Assembly	Body (see note 5)				
-1	-31	.093 (2.36)	.188 (4.78)	1.04 (26.42)	.49 (12.45)
-2	-32	.125 (3.18)	.188 (4.78)	1.09 (27.69)	.49 (12.45)
-3	-33	.203 (5.16)	.188 (4.78)	1.12 (28.45)	.55 (13.97)
-4	-34	.234 (5.94)	.188 (4.78)	1.12 (28.45)	.55 (13.97)
-7	-37	.281 (7.14)	.281 (7.14)	1.34 (34.04)	---
-9	-39	.422 (10.72)	.406 (10.31)	1.44 (36.58)	.67 (17.02)
-12	-42	.500 (12.70)	.531 (13.49)	1.75 (44.45)	.91 (23.11)
-14	-44	.656 (16.66)	.719 (18.26)	1.88 (47.75)	.94 (23.88)
-15	-45	.718 (18.24)	.719 (18.26)	1.88 (47.75)	---
-18	-48	.875 (22.23)	.938 (23.83)	2.07 (52.58)	1.13 (28.70)
-20	-50	1.093 (27.76)	1.250 (31.75)	2.18 (55.37)	1.20 (30.48)
-21	-51	1.344 (34.14)	1.500 (38.10)	2.28 (57.91)	1.27 (32.26)
-22	-52	1.813 (46.05)	1.938 (49.23)	2.46 (62.48)	1.37 (34.80)

FIGURE 1. Adapter pipe to tube - Continued.

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Non-standard sizes	Dash number	Tube OD Ends Nom inches (mm)	A Pipe thread NPTF (see note 10)	B (see note 9)	C Hex flat Nom (see note 7)
Assembly	Body (see note 5)				
-5	-35	.250 (6.35)	.250 - 18	.4375 - 20UNF	9/16
-6	-36	.313 (7.95)	.250 - 18	.5000 - 20UNF	9/16
-8	-38	.500 (12.70)	.250 - 18	.7500 - 16UNF	13/16
-10	-40	.625 (15.88)	.375 - 18	.8750 - 14UNF	15/16
-11	-41	.375 (9.53)	.500 - 14	.5625 - 18UNF	7/8
-13	-43	.750 (19.05)	.500 - 14	1.0625 - 12UN	1 1/8
-16	-46	.875 (22.23)	1.000 - 11 1/2	1.1875 - 12UN	1 3/8
-17	-47	1.000 (25.40)	.750 - 14	1.3125 - 12UN	1 3/8
-19	-49	1.250 (31.75)	1.000 - 11 1/2	1.6250 - 12UN	1 11/16

Non-standard sizes	Dash number	D dia. Drill	D_1 dia. Drill	L inches (mm) ±.02 (0.51)	S max inches (mm)
Assembly	Body (see note 5)				
-5	-35	.203 (5.16)	.281 (7.14)	1.32 (33.53)	.69 (17.53)
-6	-36	.234 (5.94)	.281 (7.14)	1.32 (33.53)	.69 (17.53)
-8	-38	.422 (10.72)	.281 (7.14)	1.44 (36.58)	.67 (17.02)
-10	-40	.500 (12.70)	.406 (10.31)	1.56 (39.62)	.813 (20.65)
-11	-41	.281 (7.14)	.531 (13.49)	1.60 (40.64)	.91 (23.11)
-13	-43	.656 (16.66)	.531 (13.49)	1.88 (47.75)	.91 (23.11)
-16	-46	.718 (18.24)	.938 (23.83)	2.07 (52.58)	1.13 (28.70)
-17	-47	.875 (22.23)	.719 (18.26)	1.88 (47.75)	.91 (23.11)
-19	-49	1.093 (27.76)	.938 (23.83)	2.15 (55.50)	.90 (22.86)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked, stored, or issued.
6. At manufacturers option, through passages may conform with the smaller diameter specified, or the appropriate end may be counterbored to the larger diameter for depth “S”.
7. Across flat widths must fit standard wrench openings, see ASME B18.2.2.
8. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
9. ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Forms)
10. ASME B1.20.3 – Dry seal Pipe Threads, Inch

FIGURE 1. Adapter pipe to tube - Continued.

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REQUIREMENTS:

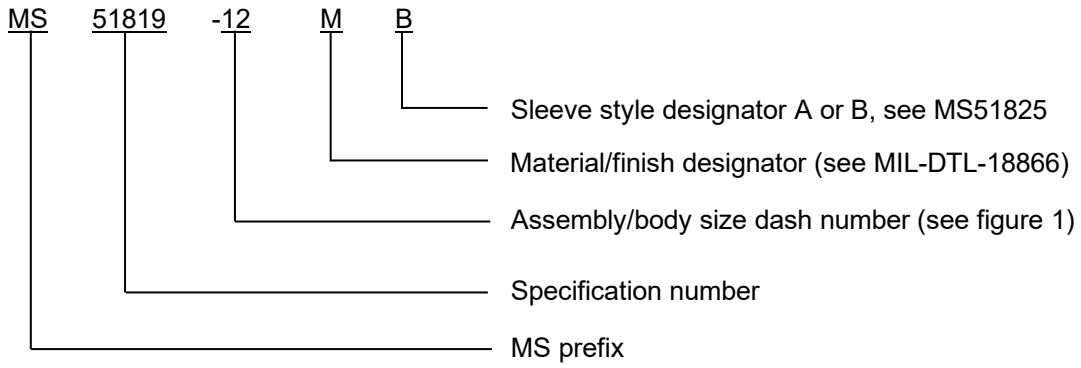
Adapter pipe to tube shall be as specified on figure 1.

Materials and finish designators shall be in accordance with MIL-DTL-18866.

Unless otherwise specified in MIL-DTL-18866, all platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Maximum operating pressure shall be in accordance with SAE-J514.

Part or Identifying Number (PIN): The PIN consists of the letters “MS”, the specification number, a dash, number for adapter assembly/tube size, a letter for material finish designator and a letter for a sleeve designator.



PIN example: MS51819-12MB indicates an adapter pipe to tube assembly, .6250 inch (15.875 mm) tube OD, to 1/2 NPTF, Nickel-copper alloy, and B style sleeve.

Marking shall be in accordance with MIL-DTL-18866.

Class I and II ozone depleting substances (ODS) shall not be used in MS51819 or any referenced procedures.

Tapered pipe thread connections are generally prohibited in high shock applications.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

NOTE: The following restrictions do not apply to manufacturers of the fittings and is applicable only to OEM’s and users.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

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Army and Marine ground vehicle applications using zinc plating. PIN codes FC, FD, Z, and ZC finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army and Marine ground systems. The use of coatings depositing hexavalent chromium for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement. For new designs PIN codes FC, FD, Z, and ZC are not to be supplied to Army and Marine ground forces. PIN code KA (MIL- PRF- 32647 Type K20) is intended as a hazardous free plating material and is a direct replacement for zinc plating with a hexavalent chromium conversion coating. PIN code KA has equivalent torque/tension behavior as the zinc plating it replaces.

Army and Marine ground vehicles. The use of cadmium plating for Army and Marine ground vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.

Army and Marine ground vehicle applications using cadmium plating. Pin code J or KB may be substituted for cadmium (“Blank” PIN code) for Army ground vehicle applications. PIN code KB (MIL-PRF-32647 Type K16) is intended as a hazardous free plating material and is a direct replacement for cadmium plating (“Blank” PIN code). PIN code KB has an equivalent torque/tension behavior as the cadmium plating it replaces.

NAVSEA shipboard owned equipment. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Table II provides a detail reference data listing for MS39206 to MS51819. This table contains coordination of equivalent tube sizes, in both compression and hydraulic type adapters. Assembly replacement is authorized.

TABLE II. Assembly equivalency (for information only).

Compression MS39206	Hydraulic MS51819	Compression MS39206	Hydraulic MS51819	Compression MS39206	Hydraulic MS51819
-1	-1	-8	-8	-15	-15
-2	-2	-9	-9	-17	-16
-3	-3	-10	-10	-16	-17
-4	-4	-11	-11	-18	-18
-5	-5*	-12	-12	-19	-19
-6	-6*	-13	-13	-20	-20
-7	-7	-14	-14	-21	-21

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MIL-PRF-32647
MS39206
MS51823
MS51825
ASME B1.1
ASME B18.2.2
ASME B1.20.3
ASTM B117
SAE-J514
T9070-AL-DPC-020/077-2

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2022-011)

Review activities:
Army - AR, MI
Navy - AS, CG, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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